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“IN MY OPINION . . .”

PARALLELS BETWEEN OLD-GROWTH FOREST AND WILDLIFE POPULATION MANAGEMENT

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There are similarities between forest and wildlife management. Both have roots in a utilitarian past that emphasized resource management for production of a sustained crop. Both harvest regimes are based on a logistic growth curve that describes the change in resource abundance through time and a point of maximum sustained yield.

Because of these parallels, I reviewed selected issues from The Wildlife Society's position statement on management and conservation of old-growth forests in the United States (The Wildl. Soc. 1988) and from The Wildlife Society's ad hoc committee on old-growth forests (Thomas et al. 1988). In this paper, I make comparisons between these forest management issues and analogous issues pertaining to big game management (Table 1). Comparisons are equally applicable to most game species, but I concentrated on big game because, as a group, big game species are longer-lived than most other game animals and we often advance a position of exploitation based on a need for management.

Clearly, significant differences exist between the 2 management efforts. Old-growth issues relate to ecosystems and functional relationships at that level, whereas big game management applies to population dynamics and behavioral interactions. Also, recovery to unharvested conditions may take centuries for forests and decades for big game populations. Despite these substantial differences, similarities should be considered. I make these com-

parisons to generate discussion within the wildlife profession.

ARGUMENTS FOR OLD-GROWTH MANAGEMENT

Old-growth forests are rare (<5% of original forests remain in the United States [The Wildl. Soc. 1988]), as are un hunted big game populations. On public lands, un hunted big game populations may be even less prevalent because old-growth forests are preserved in national parks, national forests, and designated wilderness areas, but un hunted big game populations exist only in national parks and some wildlife refuges.

Old-growth forests function differently from managed stands (The Wildl. Soc. 1988), just as un hunted populations function differently from hunted populations. Several examples are presented in the wildlife literature. Un hunted white-tailed deer (*Odocoileus virginianus*) may exhibit territoriality (Gavin et al. 1984), which is observed in hunted populations only at very high densities, if at all. Dusek et al. (1989) postulated that white-tailed deer populations could be regulated by social interactions, a phenomenon difficult to document in hunted populations. Un hunted big game animals readily habituate to human presence as evidenced by responses of grizzly bears (*Ursus arctos*) in and adjacent to Glacier National Park (McLellan and Shackleton 1989). Social interactions in mountain lion (*Felis concolor*)

Table 1. Comparisons of issues surrounding old-growth forest conservation and management of unhunted game populations.

Old-growth forests	Unhunted game populations
Rare (<5% remain) environments that function differently from managed stands. They provide high intrinsic value and extraordinary aesthetic qualities.	Rare populations that function differently from hunted populations, and possess high intrinsic value and extraordinary aesthetic qualities.
Characterized by high biomass and low net biomass accumulation.	Characterized by high density and low net productivity.
Economic value of old growth is substantial. Many local and some regional economies tied to timber harvest as is management funding (Knutson-Vandenberg). This results in political pressure to continue harvest.	Economic value of hunting is substantial. Many local and some regional economies tied to hunting season as is management funding (Pittman-Robertson). This results in political pressure to continue harvest.
Cannot be recreated with current silvicultural techniques.	Cannot be recreated with current harvest strategies.
Managers should define, inventory, and map; preserve sufficient amount and distribution to ensure unique ecological functions are maintained; provide widest array of future management options; and expand research.	Same for game populations.
Old-growth forests are unproductive, diseased, represent potential insect and disease reservoirs, threaten adjacent forests, and are a waste of a valuable resource.	Unhunted wildlife populations at carrying capacity are unproductive, suffer from malnutrition and disease, damage their habitat, and are a waste of a valuable resource.
Fire suppression has altered natural communities. Old-growth stands do not reflect pristine conditions and require management.	Predator eradication and control has altered big game populations and wildlife communities. Unhunted populations do not reflect pristine conditions and require management.

populations may be disrupted by hunting, causing increased fighting among males (Logan et al. 1986). Harvest also disrupts social interactions in grizzly bear populations that can result in decreased cub survival (Wielgus 1992). Other examples could be cited. I submit that unhunted big game populations also have high intrinsic value and extraordinary aesthetic qualities as proposed for stands of old-growth forest (Thomas et al. 1988).

Old-growth forests are characterized by high biomass and low net biomass accumulation (The Wildl. Soc. 1988). These characteristics are directly comparable to the characteristics presumed to be associated with wildlife populations at "K-carrying-capacity." Sustained yield theory suggests that unhunted big game populations should maintain high total biomass (or

density) and low net productivity compared to hunted populations maintained below carrying capacity (McCullough 1987). One important difference between forest and wildlife management may be that decreased net productivity is more directly dependent on total biomass in forests and, in fact, does not seem to function in big game management until populations closely approach "K" (Fowler 1981).

Old-growth forests are especially valuable for timber (The Wildl. Soc. 1988). Many local and some regional economies are tied to timber management (Thomas et al. 1988). The economic value of hunting also is substantial, especially in rural areas. Graham (1988) estimated total expenditures for elk (*Cervus elaphus*), deer (*Odocoileus* spp.), and prong-

horn antelope (*Antilocapra americana*) hunting in Montana during 1985 at \$127 million; net economic value represented an additional \$95 million. Both timber and wildlife management efforts support local economies and in turn are supported by funding generated from each management effort. The Pittman-Robertson Federal Aid in Wildlife Restoration Act of 1937 generates revenues used to support game management just as the Knutson-Vandenberg Act of 1930 generates funds for reforestation and management of forest resources. All these factors combine to result in political pressure to continue harvests of both timber and wildlife.

The complexity of old-growth forest ecosystems cannot be recreated with current silvicultural techniques (The Wildl. Soc. 1988). Neither have unhunted populations been successfully emulated using current big game harvest strategies. Both forest and wildlife professionals have developed harvest management programs that attempt to preserve some aspects of unmanaged stands or populations but neither has successfully duplicated complex natural conditions. For example, biological diversity initiatives in forestry strive to maintain some of the complexity found in old-growth forests through maintenance of structural diversity in managed stands (Thomas et al. 1988). Antler-point restrictions for deer and elk represent similar attempts by wildlife managers to maintain a component of mature males in harvested big game populations. A management strategy emphasizing harvest of juvenile animals (Demarchi and Woltersen 1991) is another example of wildlife managers attempting to emulate more natural population processes.

The Wildlife Society's policy statement identifies needs to: (1) define, inventory, and map existing stands of old growth; (2) preserve sufficient amounts and distribution of old growth to ensure that unique ecological functions are maintained; (3) provide the widest array of management options for the future;

and (4) expand research to gain more information about old-growth ecosystems (The Wildl. Soc. 1988). It seems these arguments are directly applicable to management of unhunted wildlife populations as well.

Several arguments have been raised in opposition to old-growth management and each has direct parallels to wildlife management. Statements such as unmanaged stands (or populations) are unproductive (i.e., nothing to harvest), are reservoirs for insects and diseases (unhunted populations may suffer from malnutrition and disease), are a threat to adjacent forests because of insects, disease, and fire (unhunted populations may damage their habitat), and are a "waste" (no human use) of a valuable resource are some examples.

Arguments regarding the impacts of past mismanagement that have destroyed pristine conditions also have direct corollaries between forest and wildlife management. For example, fire suppression efforts during most of this century have altered forest communities, just as predator control and eradication programs have altered big game populations and communities. Discussions regarding edge effects of adjacent harvested areas and the size of reserve that is needed also are comparable. Analogous arguments could be developed for either forest or wildlife management supporting the need for unharvested areas that provide baseline conditions to which managed areas can be compared.

FUTURE DIRECTION OF WILDLIFE MANAGEMENT

The wildlife profession, through The Wildlife Society, has accepted a stronger role in conservation policy debates. We have come to realize that whereas biological concerns continue to be a factor, decisions are more frequently being made in courts and public referenda, outside of agency jurisdictions. Wildlife professionals can play an extremely important role in this process by providing accurate and

balanced information to governmental agencies, other decision-making bodies, and the general public so that informed decisions are made. We must assure that this information is "unencumbered by politics of governmental agencies or the emotion of special human interests" (Mackie 1991:29). We must be equally critical in reviewing our profession even though challenging our own paradigms may be difficult.

Public attitude surveys have highlighted changing societal values. Kellert (1980) estimated that hunting opponents outnumber active hunters. Only 25% of the United States population (excluding Hawaii) had hunted at some time in their lives. Of those who had hunted, 53% no longer hunt (only 14% had hunted in the previous 2 years) and 19% of those that stopped listed personal opposition to hunting as the reason for quitting. In contrast, 25% reported that they had participated in bird watching in the previous 2 years.

Other trends project a decreasing emphasis on harvest of wildlife and a diversification of wildlife management strategies. Bergstrom and Cordell (1991) estimated that in 1987, the net economic value of big game hunting in the United States was \$1.6 billion and the value of wildlife observation was \$3.2 billion. Walsh et al. (1989) projected that nonconsumptive wildlife use will more than double from 1980–2040. Over the same period, the number of people hunting big game should remain relatively stable, resulting in a declining proportion of people participating in big game hunting.

Calls for change have emanated from within both the wildlife and forestry professions. Heberlein (1991) suggested that it is not a question of if the direction of wildlife management will change, only of when. Kania and Conover (1991) have argued that wildlife management agencies need to adapt to societal changes rather than resisting them. McQuillan (1992), discussing new perspectives in forestry, argued that society has entered a new era characterized by a concern for diversity and complexity, and turning away from a utilitarian past.

The arguments developed by The Wildlife Society on management and conservation of old-growth forests are convincing; there clearly is a need for conservation of unharvested timber reserves that extends beyond species preservation. The similarities between forest and wildlife management also require us to ask the question, is there a comparable need for establishment of unharvested "game" populations to be maintained for their intrinsic worth? Such areas could provide many of the benefits to designated game populations that have been proposed for old-growth forest reserves and would move the profession toward meeting the growing public demands for alternate uses of wildlife. In my view, the parallels between forest and wildlife management demand that we consider the implications of these arguments. Otherwise, we may find ourselves following or resisting, rather than leading, changing wildlife management policies.

SUMMARY

Many similarities exist between forest and wildlife management. Because of these parallels, I reviewed issues presented in The Wildlife Society's position statement on old-growth forest conservation and compared them to big game harvest management strategies. The issues are intriguing and many are easily transferred from forest ecosystems to big game populations. If we acknowledge the need for old-growth forest reserves, then it seems we may need to reconsider our big game harvest programs. The wildlife profession, through The Wildlife Society, has adopted a new role to provide objective, factual information to governmental leaders and other decision-makers, unencumbered by politics of governmental agencies or special interest groups. To meet this objective we must be willing to challenge our paradigms and seriously consider a changing role for wildlife managers. I encourage discussion of these issues.

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